

# SPECTER AI

## Setup & Configuration Guide

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*Your AI. Your Rules. Zero Cloud.*

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This guide walks you through installing, configuring, and running SPECTER AI on your own hardware.  
100% local - no cloud, no subscriptions, no data leaving your machine.

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## 1. What You Receive

- SPECTER Core - Voice, vision, personality, memory engine
- SOVEREIGN - Terminal AI with ReAct loops and vector memory
- Docker Compose Stack - One-command full deployment
- Config Templates - .env.example and YAML configs
- Architecture Docs - System design overview

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## 2. System Requirements

### Minimum

- OS: Ubuntu 22.04+ / Debian 12+ / macOS 13+
- RAM: 16GB min (32GB recommended)
- GPU: NVIDIA RTX 3060 8GB VRAM
- Python: 3.11+ | Docker: 24.0+
- Storage: 50GB free

### Recommended

- GPU: NVIDIA RTX 3090/4090 (24GB VRAM)
- RAM: 32-128GB | NVMe SSD

*Note: CPU-only works but voice/vision will be slow. GPU strongly recommended.*

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## 3. Prerequisites

### Install Ollama

```
curl -fsSL https://ollama.ai/install.sh | sh
ollama --version
```

### Install Docker

```
curl -fsSL https://get.docker.com | sh
sudo usermod -aG docker $USER && newgrp docker
```

### Python 3.11+

```
sudo apt-get install -y python3.11 python3.11-venv python3-pip
brew install python@3.11 # macOS
```

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## 4. Installation

```
unzip specter-ai-framework-v1-x9k3p7.zip -d ~/specter
cd ~/specter/SPECTER-AI-Framework
python3 -m venv venv && source venv/bin/activate
pip install --upgrade pip && pip install -r requirements.txt
cp .env.example .env
```

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## 5. Environment Configuration (.env)

### Core

```
SPECTER_MODEL=llama3.1:8b
OLLAMA_URL=http://localhost:11434
OPERATOR_NAME=YourName
LOG_LEVEL=INFO
```

### Voice

```
VOICE_ENABLED=true
WAKE_WORD=specter
STT_ENGINE=whisper
WHISPER_MODEL=base
TTS_ENGINE=pyttsx3
```

### Vision

```
VISION_ENABLED=true
OCR_ENGINE=tesseract
SCREEN_CAPTURE_INTERVAL=0
```

### Memory

```
QDRANT_URL=http://localhost:6333
MEMORY_COLLECTION=my_specter_memory
MEMORY_TOP_K=5
```

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## 6. Choosing Your AI Model

- 8GB VRAM: llama3.1:8b, mistral:7b, qwen2.5:7b
- 16GB VRAM: gemma2:9b, deepseek-r1:8b
- 24GB VRAM: llama3.1:70b Q4, mixtral:8x7b
- CPU Only: llama3.2:3b, phi3:mini (slower)

```
ollama pull llama3.1:8b
ollama list
```

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## 7. Launching SPECTER

```
cd ~/specter/SPECTER-AI-Framework
docker compose up -d
docker compose ps
docker compose logs -f specter
docker compose down # to stop
```

- Web Dashboard: <http://localhost:3000>
- API: <http://localhost:8000>
- Qdrant UI: <http://localhost:6333/dashboard>

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## 8. Voice Setup

```
# Find your microphone index
python3 -c "import sounddevice as sd; print(sd.query_devices())"
# Set AUDIO_DEVICE_INDEX=N in .env
```

### Install Whisper + Audio

```
pip install openai-whisper sounddevice pyaudio
sudo apt-get install -y portaudio19-dev
```

### Install Tesseract (OCR)

```
sudo apt-get install -y tesseract-ocr # Ubuntu
brew install tesseract                # macOS
```

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## 9. Creating Your Own Persona

Edit config/persona.yaml to define your AI - name, personality, expertise, communication style:

```
name: "ARIA"

identity: |
  You are ARIA, a capable AI assistant.
  Direct, analytical, and efficient.

traits:
  - analytical
  - direct
  - resourceful

communication_style: "concise and precise"

expertise:
```

- software development
- research

wake\_word\_response: "Online. Ready."

- Keep identity under 200 words for best performance
  - 3-5 traits works best
  - Test with simple questions first before complex tasks
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## 10. SOVEREIGN Terminal Agent

```
cd ~/specter/SPECTER-AI-Framework/SOVEREIGN
pip install -r requirements.txt
cp config/sovereign.yaml.example config/sovereign.yaml
python3 -m sovereign
```

- ReAct Loop - Multi-step autonomous reasoning
- Tool Registry - Bash, file ops, code execution
- Vector Memory - Persists across sessions via Qdrant
- Cost Tracker - Token usage monitoring
- TUI Dashboard - Terminal UI with status panels

### Example Commands

```
> Analyze ~/project files and summarize each one
> Write a script monitoring CPU every 5 seconds
> Remember my server IP is 192.168.1.100
> What do I know about machine learning?
```

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## 11. Memory System (Qdrant)

Docker Compose starts Qdrant automatically. To start manually:

```
docker run -d --name qdrant -p 6333:6333 qdrant/qdrant
curl http://localhost:6333/health
```

- Say: Remember that [fact] - stores information
  - Say: What do you know about [topic] - retrieves
  - Say: Forget everything about [topic] - deletes
  - Memory persists automatically between sessions
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## 12. Troubleshooting

## Cannot connect to Ollama

```
curl http://localhost:11434/api/tags  
ollama serve
```

## No audio / mic errors

```
sudo apt-get install -y portaudio19-dev  
pip install sounddevice pyaudio
```

## GPU not detected

```
nvidia-smi  
docker run --gpus all nvidia/cuda:12.0-base nvidia-smi
```

## Out of VRAM

- ollama pull llama3.1:8b-q4\_0 (quantized model)
- Set WHISPER\_MODEL=tiny in .env
- Close other GPU-intensive applications

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## 13. Support

- Email: oblivion.designz@gmail.com
- Include: system specs + docker compose logs specter
- Response within 24 hours
- Follow @OblivionLabz on X for updates
- Check oblivionlabz.net for new releases